

```
!pip install gensim
```

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Collecting gensim
```

```
  Downloading gensim-4.4.0-cp312-cp312-  
manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8.4 kB)  
Requirement already satisfied: numpy>=1.18.5 in  
/usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)  
Requirement already satisfied: scipy>=1.7.0 in  
/usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)  
Requirement already satisfied: smart_open>=1.8.1 in  
/usr/local/lib/python3.12/dist-packages (from gensim) (7.5.0)  
Requirement already satisfied: wrapt in  
/usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1-  
>gensim) (2.0.1)  
Downloading gensim-4.4.0-cp312-cp312-  
manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9 MB)  
27.9/27.9 MB 16.8 MB/s eta  
0:00:00
```

```
Successfully installed gensim-4.4.0
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```
import numpy as np  
import gensim.downloader as api
```

```
model = api.load("word2vec-google-news-300")
```

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[=====] 100.0%  
1662.8/1662.8MB downloaded
```

```
def association(w, A, B, model):  
    return np.mean([model.similarity(w, a) for a in A if a in model])  
- \  
    np.mean([model.similarity(w, b) for b in B if b in model])
```

```
def weat_score(X, Y, A, B, model):  
    return np.sum([association(x, A, B, model) for x in X if x in  
model]) - \  
    np.sum([association(y, A, B, model) for y in Y if y in  
model])
```

```
def effect_size(X, Y, A, B, model):  
    scores = [association(w, A, B, model) for w in X + Y if w in  
model]  
    mean_x = np.mean([association(x, A, B, model) for x in X if x in  
model])  
    mean_y = np.mean([association(y, A, B, model) for y in Y if y in  
model])  
    return (mean_x - mean_y) / np.std(scores)
```

```
male_words = ["man", "male", "boy", "brother", "he", "him", "son"]  
female_words =
```

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["woman", "female", "girl", "sister", "she", "her", "daughter"]

career_words =
["executive", "management", "professional", "corporation", "salary", "office", "career"]
family_words =
["home", "parents", "children", "family", "cousins", "marriage", "wedding"]

christian_words = ["church", "bible", "christian"]
muslim_words = ["mosque", "quran", "muslim"]

positive_words = ["honest", "kind", "peace", "love"]
negative_words = ["terror", "violence", "hate", "anger"]

gender_weat = weat_score(male_words, female_words, career_words,
family_words, model)
gender_effect = effect_size(male_words, female_words, career_words,
family_words, model)

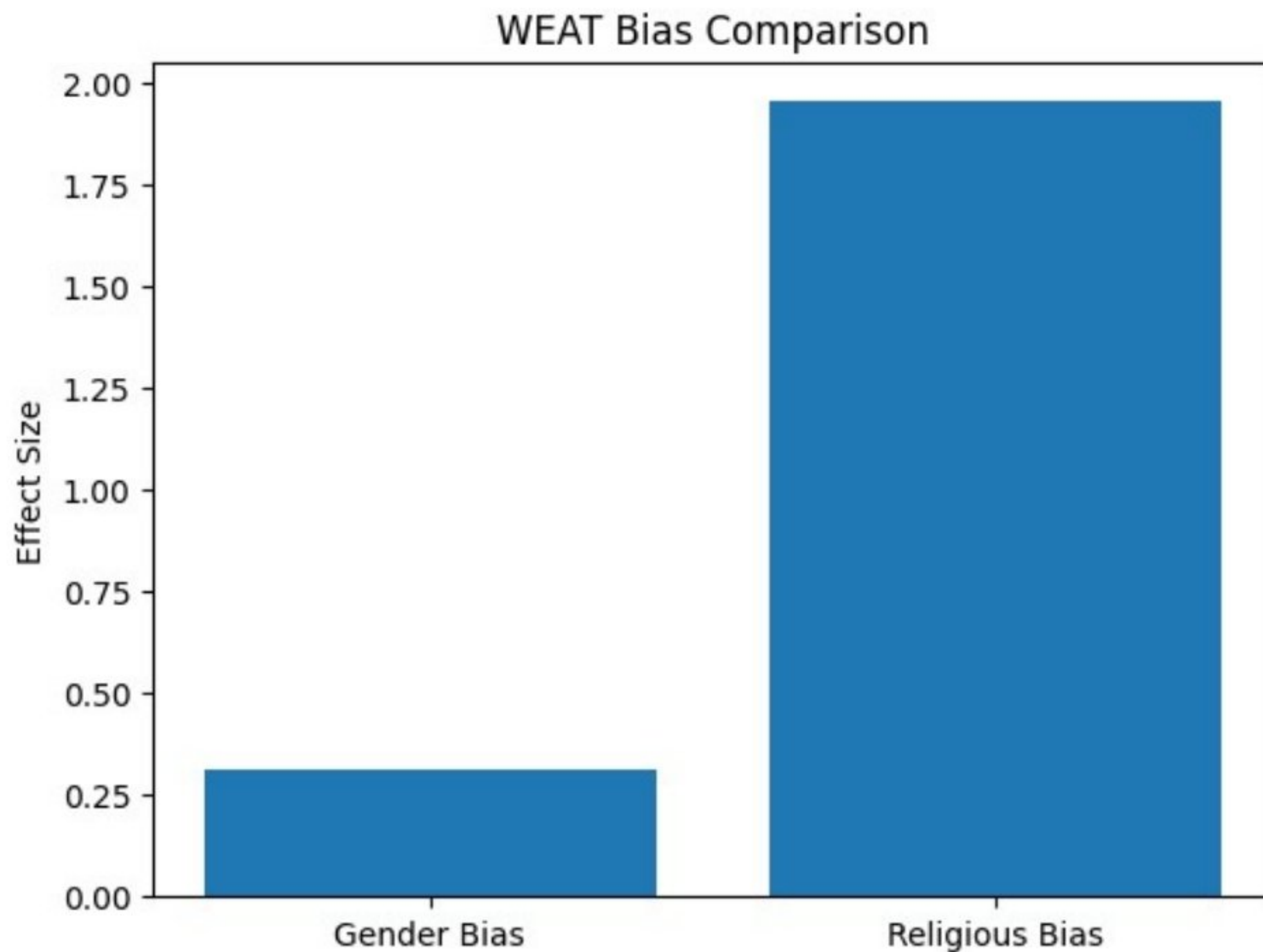
religious_weat = weat_score(christian_words, muslim_words,
positive_words, negative_words, model)
religious_effect = effect_size(christian_words, muslim_words,
positive_words, negative_words, model)

effects = [gender_effect, religious_effect]
labels = ["Gender Bias", "Religious Bias"]

import matplotlib.pyplot as plt

plt.bar(labels, effects)
plt.ylabel("Effect Size")
plt.title("WEAT Bias Comparison")
plt.show()

```



```
print(f"Gender WEAT Score: {gender_weat:.4f}")
print(f"Gender Effect Size: {gender_effect:.4f}")

print(f"\nReligious WEAT Score: {religious_weat:.4f}")
print(f"Religious Effect Size: {religious_effect:.4f}\n")

if gender_effect > 0:
    print("Bias Direction: Career is more associated with MALE terms")
elif gender_effect < 0:
    print("Bias Direction: Career is more associated with FEMALE terms")
else:
    print("No detectable gender bias")

if religious_effect > 0:
    print("Bias Direction: Positive words are more associated with CHRISTIAN terms")
elif religious_effect < 0:
    print("Bias Direction: Positive words are more associated with MUSLIM terms")
else:
    print("No detectable religious bias")
```

Gender WEAT Score: 0.2905
Gender Effect Size: 0.3136

Religious WEAT Score: 0.2987
Religious Effect Size: 1.9514

Bias Direction: Career is more associated with MALE terms
Bias Direction: Positive words are more associated with CHRISTIAN terms